Fiscal Year:	FY 2004	**Task Last Updated:**	FY 12/02/2004
PI Name:	Stroud, Kenneth J. Ph.D.		
Project Title:	Virtual Reality Training: Prevention & Prediction of Space Motion Sickness and Disorientation		
Division Name:	Human Research		
Program/Discipline:	ADVANCED HUMAN SUPPORT TECHNOLOGIES		
Program/Discipline--Element/Subdiscipline:	ADVANCED HUMAN SUPPORT TECHNOLOGIES--Space human factors engineering		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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Zip Code:	80309-0429	**Congressional District:**	2
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	GSRP
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No. of Post Docs:	0	**No. of PhD Degrees:**	0
No. of PhD Candidates:	0	**No. of Master' Degrees:**	1
No. of Master's Candidates:	1	**No. of Bachelor's Degrees:**	0
No. of Bachelor's Candidates:	0	**Monitoring Center:**	NASA JSC
Contact Monitor:		**Contact Phone:**	
Contact Email:			
Flight Program:			
Flight Assignment:			
Key Personnel Changes/Previous PI:			
COI Name (Institution):			
Grant/Contract No.:			
Performance Goal No.:			
Performance Goal Text:			
Task Description:	The overall goal of the proposed research is to determine if variable training in multiple orientations in a virtual environment will decrease motion sickness and improve task performance upon exposure to a novel orientation at a later time, as compared to nonvariable training. This study proposes to: 1) determine the subjects' orientation rest frame type as either visual-spatial (VS) or internal z-axis (IZ), 2) determine if variable practice training in different initial orientations will result in reduced motion sickness and better task performance in subjects faced with a novel initial orientation, and 3) determine if and how orientation rest frame affects motion sickness and task performance. Specific responses that will be examined include latency to the onset of vection (for classification of orientation rest frame), motion sickness symptoms and symptom severity, and task performance as measured by accuracy and time to task completion. Space Motion Sickness (SMS) and vestibular disturbances can impact space flight crew productivity and safety. The potentially debilitating effects of SMS during the initial days of flight cause certain critical operations such as docking to the International Space Station (ISS) and extravehicular activity (EVA) to be delayed for several days.		

	Research suggests that orientation rest frame, which can be determined by latency to vection (perceived self-motion) onset, may be a predictive factor for SMS. Subjects with shorter latencies have been classified as having a VS frame orientation of reference, while those with longer latencies have been classified as having a IZ frame of reference. Research also suggests that training devices, such as those using virtual reality, can partially simulate the sensory rearrangements present in microgravity. The plasticity of the human central nervous system makes it likely that subjects will eventually adapt to the novel sensory conditions present in a virtual environment. This study is designed to provide important information that will support the long range goal of developing preflight training procedures, (using virtual environments), that will allow astronauts to adapt to sensory conditions similar to microgravity before the flight occurs. Adaptation to these conditions is expected to reduce SMS, and sensorimotor and orientation disturbances on orbit.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	Enhanced understanding of terrestrial motion sickness, and possible application of study protocols for terrestrial motion sickness and disorientation prevention.
Task Progress:	During the past year of funding through the NASA Graduate Student Research Program, a number of important steps related to the research proposal have been accomplished. Most importantly, an experimental, titled “Virtual Reality Training for Spatial Orientation in Unfamiliar Environments” was completed using 40 subjects from the NASA JSC Human Test Subject Facility, to examine the ability to predict and prevent the occurrence of motion sickness and disorientation during exposure to a virtual environment similar to space flight. During the Spring of 2003, data collection was completed on 30 subjects, and upon subsequent data analysis, data from an additional 10 subjects was completed in November. The results from the first 30 subjects, which are very promising, were presented at the International Astronautical Congress in Bremen, Germany in October. The overall concept of this study and future studies was presented at the International Conference on Environmental Systems (ICES) in Vancouver, British Columbia in July. Two to three additional manuscripts will be submitted to journals summarizing the results of the complete study. In addition to examining the feasibility of using virtual reality training for predicting and preventing Space Motion Sickness and disorientation, research into spacecraft design requirements to reduce vestibular disturbances upon return from a longer mission was completed during Spring of 2004. While SMS is the primary concern for short-duration missions, the effect of vestibular disturbances during landing may be particularly problematic following long-duration missions such as returning from the ISS, or on a Mars mission, where the vestibular system would likely have adapted from 1 g, to 0 g, to 0.38 g, and then the reverse for the return trip. Due to the combination of the perceptual illusions inherent in flying, vestibular adaptation to different gravity environments, diminished muscle strength and proprioception, the many different directions of acceleration during reentry and landing, and the design of the spacecraft, vision, attitude, position, and motion perception will almost certainly be affected and could impair the astronaut’s ability to pilot the vehicle. Design aspects such as seat position and instrument and window location must be taken into account to provide a spacecraft environment to ensure a safe landing. The results from this research will be submitted to an appropriate journal.
Bibliography Type:	Description: (Last Updated: 02/27/2004)